Program: **Diploma in Cloud Computing and Big Data**	
Course Code : **3279**	Course Title: **Programming in Python Lab**
Semester : **3**	Credits: **0**
Course Category: **Program Core**	
Periods per week: **4 (L:0 T:0 P:4)**	Periods per semester: **60**

Course Objectives:

- Develop programs by utilizing the modules Lists, Tuples, Sets and Dictionaries in Python
- Implement file operations, string processing with exception handling etc using python
- Develop python programs using Functions and Object Oriented Programming
- Develop programs in Python to process data stored in files by utilizing the modules Numpy, Matplotlib, and Pandas

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basics in Python	1008	Introduction to IT systems Lab	I
Control Structures, Modular Program Design, Arrays, Functions	2131	Problem Solving and Programming	II
	2139	Problem Solving and Programming Lab	II

Course Outcomes:

On completion of the course, the student will be able to:

CO1	Description	Duration (Hours)	Cognitive Level
CO1	Solve problems by utilizing the built in modules List, Tuple, Set and Dictionary in Python program	12	Applying

CO2	Illustrate file operations, string processing with exception handling using python	12	Applying
CO3	Demonstrate Python programs using Functions and Object Oriented Programming	14	Applying
CO4	Use the modules Numpy, Matplotlib, and Pandas to process data stored in files by python programs	16	Applying
	Lab Exams	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3		3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Solve problems by utilizing the built in modules Lists, Tuples, Sets and Dictionaries in Python program**		
M1.01	Experiment with Python Data types-Numbers (int, float, complex), strings, Boolean etc to solve various arithmetic calculations.	3	Applying
M1.02	Demonstrate the use of List, nested List, List operations, iteration over List, Tuple, Tuple operations	6	Applying
M1.03	Illustrate the use of Dictionary and Set	3	Applying

CO2	**Illustrate file operations and string processing with exception handling using python**		
M2.01	Solve simple problems using python programs for opening, reading and writing text files	3	Applying
M2.02	Demonstrate String processing - String traversal, string methods	4	Applying
M2.03	Illustrate exception handling, propagation of raised exception, catching and handling exceptions. User input exception handling and file processing	5	Applying
	Lab Exam – I	3	
CO3	**Demonstrate Python programs using Functions and Object Oriented Programming**		
M3.01	Demonstrate the concepts of Functions, return values, parameter passing, keyword arguments, default arguments, variable scope	5	Applying
M3.02	Solve problems of recursive nature using recursive function in python	4	Applying
M3.03	Illustrate the concepts of Classes , fundamental features of object oriented programming- encapsulation, inheritance and polymorphism	5	Applying
CO4	**Use the modules Numpy, Matplotlib, and Pandas to process data stored in files by python programs**		
M4.01	Demonstrate the module NumPy Basics, Creating arrays, Arithmetic, Slicing, Matrix Operations and Random numbers.	6	Applying
M4.02	Illustrate with CSV files using Pandas	6	Applying
M4.03	Use Matplotlib -Basic plot for Plotting and visualization.	4	Applying
	Lab Exam – II	3	

Text /Reference:

T/R	Book Title/Author
T1	Introduction to Computer Science Using Python - A Computational Problem - Solving Focus - Chales Dierbach Wiley India Pvt. Ltd
R1	Yashavant Kanetkar, Aditya Kanetkar, “Let us Python, BPB publication, 1st Edition, 2019

Online Resources:

S.No	Website Link
1	https://docs.python.org/3/
2	https://www.programiz.com/python-programming